

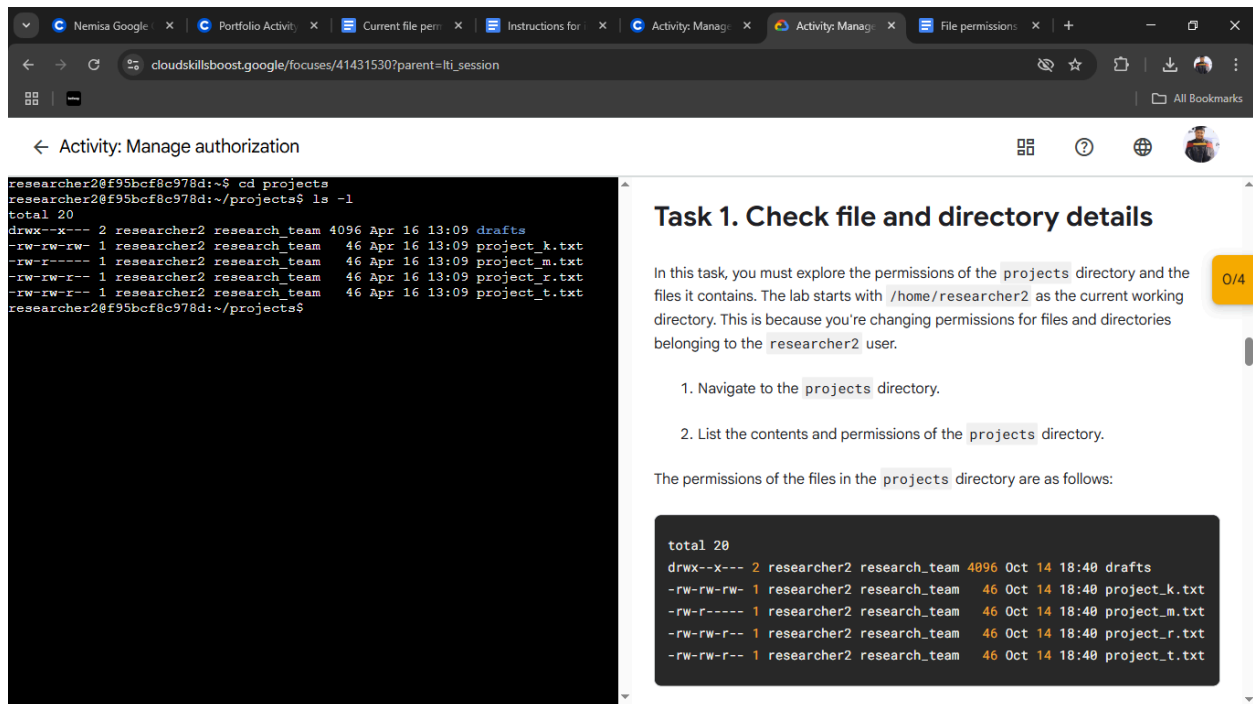
File permissions in Linux

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Project description

In this project, I managed file and directory permissions using Linux commands to ensure proper access control for a user and group. Specifically, I audited and corrected the file permissions in the `/home/researcher2/projects` directory for the `researcher2` user, who is a member of the `research_team` group. I listed and verified permissions for all files, including hidden ones, using the `ls -la` command. Where necessary, I applied the appropriate permissions using `chmod` to enforce security and access standards. Additionally, I reviewed and restricted access to the `/home/researcher2/projects/drafts` subdirectory to eliminate any unauthorized access. This task demonstrated my ability to apply Linux file system permission concepts, including user, group, and other access control, in a practical environment

Check file and directory details



Activity: Manage authorization

researcher2@f95bcf8c978d:~\$ cd projects
researcher2@f95bcf8c978d:~/projects\$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 16 13:09 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Apr 16 13:09 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Apr 16 13:09 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_t.txt
researcher2@f95bcf8c978d:~/projects\$

Task 1. Check file and directory details

In this task, you must explore the permissions of the `projects` directory and the files it contains. The lab starts with `/home/researcher2` as the current working directory. This is because you're changing permissions for files and directories belonging to the `researcher2` user.

1. Navigate to the `projects` directory.
2. List the contents and permissions of the `projects` directory.

The permissions of the files in the `projects` directory are as follows:

```
total 20  
drwx--x--- 2 researcher2 research_team 4096 Oct 14 18:40 drafts  
-rw-rw-rw- 1 researcher2 research_team 46 Oct 14 18:40 project_k.txt  
-rw-r----- 1 researcher2 research_team 46 Oct 14 18:40 project_m.txt  
-rw-rw-r-- 1 researcher2 research_team 46 Oct 14 18:40 project_r.txt  
-rw-rw-r-- 1 researcher2 research_team 46 Oct 14 18:40 project_t.txt
```

Describe the permissions string

`drwx--x--- 2 researcher2 research_team 4096 Oct 14 18:40 drafts`

- `d`: This is a **directory**.
- `rw``x`: The **owner (researcher2)** has **read, write, and execute** permissions.
- `--x`: The **group (research_team)** has **execute-only** permission, meaning members can enter the directory but cannot list or read its contents.
- `---`: **Others** have **no access** at all.
- This setup restricts directory visibility and is often used to limit access to sensitive or work-in-progress content.

`-rw-rw-rw- 1 researcher2 research_team 46 Oct 14 18:40 project_k.txt`

- `-`: This is a **regular file**.
- `rw-`: The **owner** can **read and write**.
- `rw-`: The **group** can also **read and write**.
- `rw-`: **Others** can **read and write**, which poses a security risk if not intended, as anyone can modify the file.

`-rw-r----- 1 researcher2 research_team 46 Oct 14 18:40 project_m.txt`

- `-`: Regular file.
- `rw-`: The **owner** has read and write access.
- `r--`: The **group** can only read.

- `---`: **Others** have no permissions.
- This is a safer configuration when sharing files within a group but wanting to restrict others.

```
-rw-rw-r-- 1 researcher2 research_team 46 Oct 14 18:40 project_r.txt
```

- `-`: Regular file.
- `rw-`: The **owner** has read and write access.
- `rw-`: The **group** has read and write access.
- `r--`: **Others** have read-only access.
- This allows for open viewing but restricts editing to the owner and group.

```
-rw-rw-r-- 1 researcher2 research_team 46 Oct 14 18:40 project_t.txt
```

Change file permissions

Activity: Manage authorization

```
researcher2@f95bcf8c978d:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 16 13:09 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Apr 16 13:09 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Apr 16 13:09 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_t.txt
researcher2@f95bcf8c978d:~/projects$ chmod o-w project_k.txt
researcher2@f95bcf8c978d:~/projects$
```

In this task, you must determine whether any files have incorrect permissions and then change the permissions as needed. This action will remove unauthorized access and strengthen security on the system.

None of the files should allow the other users to write to files.

1. Check whether any files in the `projects` directory have write permissions for the owner type of other.

Which file grants other users write permissions?

☐ project_m.txt

☒ project_k.txt

☐ project_t.txt

Submit

2. Change the permissions of the file identified in the previous step so that the owner type of other doesn't have write permissions.

Change file permissions on a hidden file

Activity: Manage authorization

```
researcher2@f95bcf8c978d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:09 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:43 ..
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 16 13:09 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Apr 16 13:09 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_t.txt
researcher2@f95bcf8c978d:~/projects$ chmod u-w,g+w,gr .project_x.txt
researcher2@f95bcf8c978d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:09 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:43 ..
-r--r----- 1 researcher2 research_team 46 Apr 16 13:09 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 16 13:09 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Apr 16 13:09 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_t.txt
researcher2@f95bcf8c978d:~/projects$
```

The file `.project_x.txt` is a hidden file that has been archived and should not be written to by anyone. (The user and group should still be able to read this file.)

1. Check the permissions of the hidden file `.project_x.txt` and answer the question that follows.

Which owner type has the incorrect write permissions?

☐ Just the group

☐ Just the user

☒ The user and the group

Submit

2. Change the permissions of the file `.project_x.txt` so that both the user and the group can read, but not write to, the file.

Note: Be sure to start the name of a hidden file with a period (.)

Change directory permissions

The screenshot displays a Google Cloud Skills Boost activity titled "Activity: Manage authorization". On the left, a terminal window shows the following commands and output:

```
researcher2@f95bcf8c978d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:09 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:43 ..
-r--r----- 1 researcher2 research_team 46 Apr 16 13:09 .project_x.txt
drwxr-xr-x 2 researcher2 research_team 4096 Apr 16 13:09 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Apr 16 13:09 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Apr 16 13:09 project_t.txt
researcher2@f95bcf8c978d:~/projects$ cd drafts
researcher2@f95bcf8c978d:~/projects/drafts$ ls -la
total 8
drwxr-xr-x 2 researcher2 research_team 4096 Apr 16 13:09 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:09 ..
researcher2@f95bcf8c978d:~/projects/drafts$ cd
researcher2@f95bcf8c978d:~$ cd projects
researcher2@f95bcf8c978d:~/projects$ chmod g-x drafts
researcher2@f95bcf8c978d:~/projects$ cd drafts
researcher2@f95bcf8c978d:~/projects/drafts$ ls -la
total 8
drwx----- 2 researcher2 research_team 4096 Apr 16 13:09 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 16 13:09 ..
researcher2@f95bcf8c978d:~/projects/drafts$
```

On the right, a quiz interface is shown with the following text:

group permissions of the `/home/researcher2/projects/drafts` directory and then modify the permissions as required. (You should be in the `projects` directory while managing the permissions of its subdirectory `drafts`.)

Only the `researcher2` user should be allowed to access the `drafts` directory and its contents. (This means that only `researcher2` should have execute privileges.)

1. Check the permissions of the `drafts` directory and answer the following question.

Does the group have permissions set to access the drafts directory and its contents?

☒ No
☒ Yes

Submit

2. Remove the execute permission for the group from the `drafts` directory.

Summary

In this task, I successfully audited and managed file and directory permissions within the `/home/researcher2/projects` directory. Using Linux commands such as `ls -la`, and `chmod`, I verified that all files had the appropriate access rights according to the roles of the `researcher2` user and the `research_team` group. I corrected misconfigured permissions to enhance security, such as removing write access for others on `project_k.txt` and adjusting the `drafts` directory to restrict unauthorized viewing. This exercise demonstrated my ability to apply Linux file system permissions to protect sensitive data and enforce proper access control in a collaborative environment.